Adversarial Example Detection Using Feature Consistency in Vision Transformers

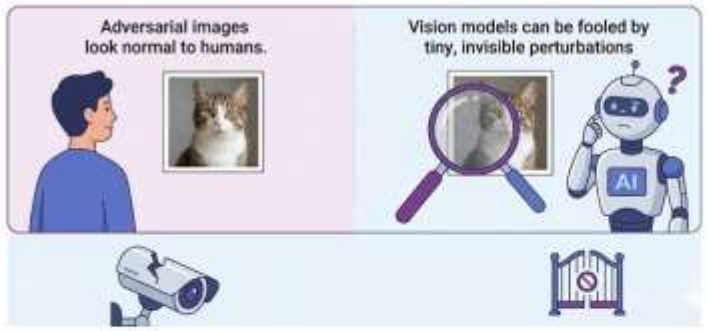
submitted by :

2005078

2005081

Why Do We Care About Adversarial Examples?

- Vision models can be fooled by **tiny, invisible perturbations**
- Adversarial images look normal to humans
- Existing detectors often fail or are easy to bypass



Key Observation

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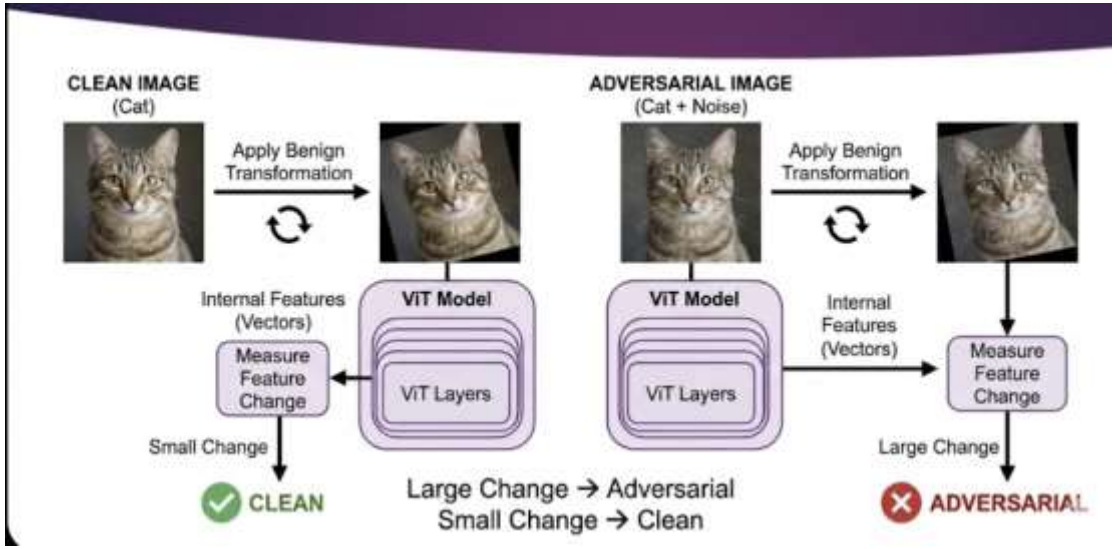
- Adversarial images are **fragile**
- Small harmless changes can break the attack
- Clean images remain **stable**
- This difference can be used for detection

Gap in Existing Work

- **What Is Missing in Existing Work?**
- Most methods look only at:
  - final prediction
  - confidence
- They do **not check internal understanding**
- Vision Transformers are largely unexplored for detection

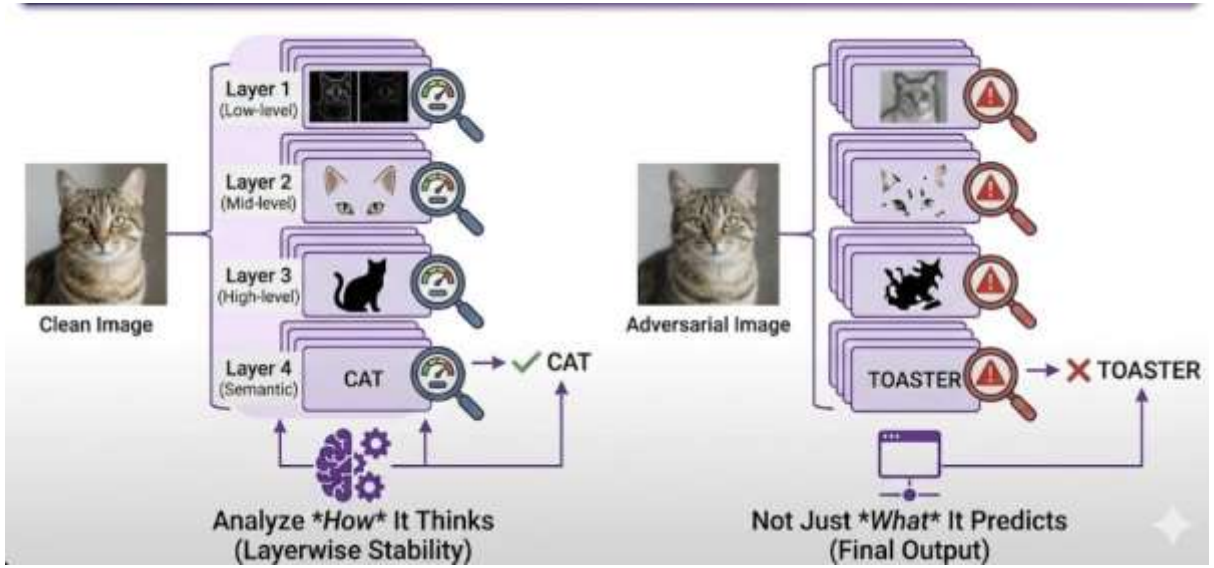
Our Core Idea

- Apply a small **benign transformation** to the image
- Extract internal features from ViT layers
- Measure how much features change
- Large change → adversarial
- Small change → clean



Why Vision Transformers ?

- Process images through **multiple layers**
- Each layer captures different understanding
- Adversarial instability appears across layers
- Enables **layerwise stability analysis**



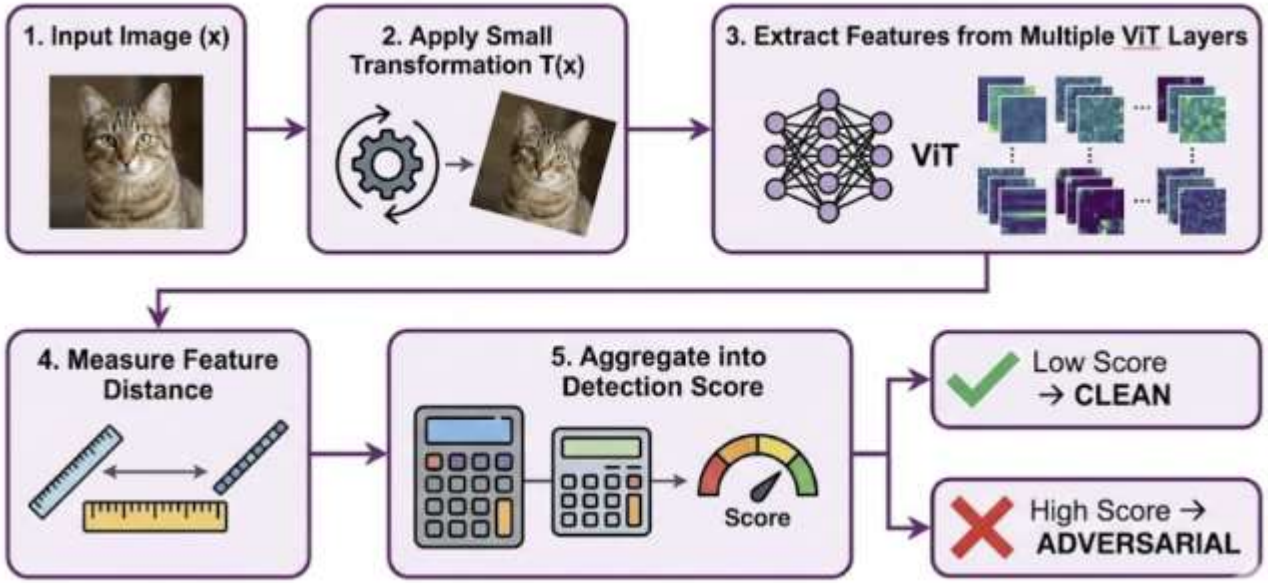
ViTs allow us to analyze *how* the model thinks, not just *what* it predicts.

Method Overview

1. Input image xxx
2. Apply small transformation T(x)T(x)T(x)
3. Extract features from multiple ViT layers
4. Measure feature distance
5. Aggregate into a detection score

▶ **Decision:**

- Low score → clean
- High score → adversarial



Contributions & Feasibility

Training-Free Detection

Uses Feature Consistency (Not Confidence)

ViT-Specific, Layerwise Analysis

Datasets: CIFAR-10 & CIFAR-100

Leverages Pretrained ViTs

Fits Available GPU Resources (Kaggle)

Achievable within Project Timeline